

SoleMate

Project Team

Ahmad Tashfeen
22I-2490 Ahmad Tariq
22I-1534
M. Umer Qureshi 22I-2578

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Supervised by

Ms. Laiba Imran



Department of Software Engineering

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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The project will be executed in four iterations, each aligned with academic evaluation milestones. Table 2.2 outlines the planned tasks and deliverables for each iteration.

Based on Comparison & Alignment with Standard Models

- ♦ Best Fit: Layered (N-tier) with Modular Services
- ♦ Why Not Pure Monolithic?
- ♦ Why Not Full Microservices?
- ♦ Why Not SOA or EDA?
- ♦ Hybrid Suggestion (Most Realistic for This Project)
- ♦ Comparative Evaluation

Final Verdict

1. Detailed analysis and recommendation for Hybrid Architecture (Layered/N-tier + selective modular/mircoservice extension)

Architecture Diagram, Layout, and Iteration Plan1) High-Level Hybrid Architecture2) Gradle Module and Package Structure3) Iteration 1 → 2 Implementation ChecklistAcceptance ExamplesDeveloper WorkflowComponent Mapping (Summary)

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Chapter 1

Introduction

This chapter introduces the project by outlining the problem being addressed, the motivation behind selecting the topic, the proposed solution, and the key stakeholders involved.

1.1 Problem Statement

With the rapid growth of e-commerce, online footwear shopping has become increasingly popular due to its convenience and accessibility. However, customers often face significant challenges when purchasing shoes virtually. The inability to physically try on footwear leads to uncertainty regarding size, comfort, and appearance, which contributes to a high rate of returns and reduced customer satisfaction. Additionally, static product images and generic sizing charts fail to capture the personalized nature of footwear choices, leaving users unable to visualize how a pair of shoes would look with their outfits or reflect their individual style.

These limitations highlight a clear gap in current e-commerce platforms, where customer experience and personalization remain underdeveloped. There is a growing demand for innovative solutions that not only enable users to virtually try on shoes in real-time but also provide features such as outfit matching, style customization, and personal history management. Addressing these challenges requires the integration of Augmented Reality (AR) and intelligent recommendation systems to bridge the gap between physical and online shopping experiences.

1.2 Motivation

The motivation for this project arises from the pressing need to enhance the online shopping experience in Pakistan and the broader Southeast Asian market. As consumer re-

liance on digital retail platforms continues to grow, the absence of reliable and interactive tools for product visualization has created a disconnect between customer expectations and the products they receive. The high rate of returns in the footwear sector not only causes financial strain for retailers but also undermines customer trust, discouraging long-term adoption of online shopping. Furthermore, the logistical burden of reverse shipments contributes to increased operational costs and environmental waste, highlighting the urgency of developing sustainable and customer-centric solutions.

The emergence of Augmented Reality (AR) provides a promising avenue to address these challenges. By enabling real-time visualization of footwear on the user's feet, AR try-on technology can significantly improve confidence in purchase decisions. Beyond fit and style validation, AR creates opportunities for personalized experiences, such as outfit matching, customized shoe skins, and digital closets that track try-on history. These features not only reduce uncertainty but also enhance customer engagement, transforming the shopping journey into an immersive and interactive experience. For academic and research purposes, this project represents an opportunity to apply computer vision, AR frameworks, and intelligent recommendation systems to a practical, real-world problem, while for industry it demonstrates a scalable pathway to improving customer satisfaction and reducing unsustainable return rates. As demonstrated in prior research, AR try-on applications can significantly reduce uncertainty in online shopping [IRJET, 2024].

1.3 Problem Solution

To address the challenges identified in the problem statement and motivation, this project proposes an Augmented Reality (AR)-based solution with the following objectives and features:

1.3.1 AR-based Real-Time Try-On

Develop an ARCore-powered virtual shoe try-on feature that overlays realistic 3D shoe models onto the user's feet in real time. The system will use computer vision (object recognition, 3D modeling) and motion tracking to align digital footwear with the user's foot movements [IRJET, 2024].

1.3.2 Outfit Matching Engine

Implement an outfit recommendation subsystem that analyzes the user's current attire

and suggests complementary items. The app will extract visual features (color) from the user's

outfit and compute similarity to other clothing/shoe options, thereby generating cohesive outfit suggestions [Statso, 2024].

1.3.3 Personalized Shoe Customization

Provide tools for customizing shoe “skins” (appearance). Users will be able to change attributes such as color, pattern, and texture and immediately preview designs in AR. This customization allows users to virtually “wear” their custom designs, engaging them creatively in the process [?].

1.3.4 Try-On History (Digital Closet)

Maintain a persistent record of all shoes and outfits a user has virtually tried on. This virtual “closet” enables easy revisit and re-try of past choices. By leveraging this history and preferences, the system can personalize future recommendations, boosting engagement and satisfaction [BrandXR, 2024].

1.3.5 Intuitive User Interface

Design a user-friendly interface with clear controls for browsing shoes, initiating AR try-on, and applying customizations. Smooth and responsive AR interactions will improve the shopping experience, allowing users to switch views, select new items, or adjust designs confidently [BrandXR, 2024].

1.3.6 Performance Optimization

Optimize the app for Android devices by using high-fidelity 3D assets and efficient rendering/tracking algorithms. Smooth, low-latency AR performance is essential for immersion and user trust [BrandXR, 2024].

1.3.7 Catalog Integration

Integrate an extensive shoe catalog so that users can browse and switch among models during try-on. Linking the AR system to a product database supports seamless exploration of retailer inventory and aligns the app with e-commerce systems [BrandXR, 2024].

1.4 Stakeholders

The success of the proposed system depends on multiple stakeholders, each playing a distinct role in its design, implementation, usage, and evaluation:

- **End Users (Online Shoppers):** Primary stakeholders who will interact with the mobile application to virtually try on shoes, customize designs, and explore outfit recommendations. Their feedback and satisfaction determine the overall success of the system.
- **Retailers and E-Commerce Platforms:** Secondary stakeholders who may integrate the system into their platforms to reduce return rates, improve customer engagement, and enhance sales conversions. These stakeholders become especially relevant when integration with commercial e-commerce applications is pursued.
- **Development Team (FYP Members):** Responsible for the system's design, implementation, testing, and deployment. Each member contributes to specific modules such as AR integration, UI/UX design, database management, and recommendation systems.
- **Supervisors and Academic Evaluators:** Provide guidance, technical oversight, and project evaluation. They ensure that the project meets academic standards and contributes meaningfully to research objectives.

Chapter 2

Project Description

This chapter provides a detailed description of the proposed project, including its scope, modular structure, tools and technologies, division of responsibilities among team members, and the planned timeline for development.

2.1 Scope

The scope of this project encompasses the design and development of a mobile-based Augmented Reality (AR) application aimed at transforming the online shoe shopping experience for customers in Pakistan and the broader Southeast Asian region. The core functionality of the system is to provide users with a realistic AR-based shoe try-on feature, enabling them to virtually visualize how a selected pair of shoes would look when worn on their feet. By integrating advanced foot tracking and overlay techniques, the application ensures accurate alignment and realistic visualization, thereby reducing uncertainty in purchase decisions. In addition to virtual try-on, the application will incorporate an outfit matching recommendation system that suggests shoes based on clothing colors and styles provided by the user, enhancing personalization and decision-making. Another integral module is the customizable shoe skin feature, which allows users to experiment with different textures, colors, and patterns on shoe models, offering a degree of personalization not typically available in conventional e-commerce platforms. The application will also maintain a try-on history, enabling users to revisit previously tried shoe models for comparison and convenience.

The system will be developed primarily for the Android platform to ensure accessibility for the majority of the regional target audience. While the primary focus is on functionality, the application will be designed with an intuitive and user-friendly interface to ensure seamless interaction for both tech-savvy and non-technical users. The solution will leverage a combination of AR frameworks, computer vision techniques, and mobile de-

velopment tools to achieve optimal performance across a wide range of Android devices. To ensure relevance and scalability, the project will be designed with modularity in mind, allowing future integration of additional features such as social sharing, AI-driven style recommendations, and brand-specific sizing guides.

The scope explicitly covers mobile AR-based visualization, user personalization features, and recommendation systems, while excluding back-end e-commerce infrastructure such as order placement, payment processing, and logistics management. Instead, the application will be developed as a complementary tool that can be integrated with existing e-commerce platforms. This project therefore positions itself as a novel, customer-centered AR solution aimed at reducing product return rates, improving buyer confidence, and enhancing customer satisfaction in the online footwear retail sector.

2.2 Modules

The proposed system is structured into multiple modules, each responsible for delivering distinct features that collectively enhance the online footwear shopping experience. The modules are grouped into **Core Modules**, which form the foundation of the system, and **Additional Features**, which extend functionality and improve user experience.

2.2.1 Core Modules

2.2.1.1 AR Try-On Module

1. Static shoe overlay on user's foot (initial prototype).
2. Scaling and rotation of shoe models to fit foot orientation.
3. Real-time AR alignment with basic foot tracking.
4. Markerless AR visualization for a natural experience.

2.2.1.2 Outfit Matching Module

1. Extract dominant outfit colors from uploaded images.
2. Suggest shoes based on complementary color schemes.
3. Rule-based style matching (e.g., sneakers for casual, formal shoes for office wear).
4. Basic catalog filtering by outfit type.

2.2.1.3 Personalized Shoe Skins Module

1. Provide 2D shoe templates (texture unwrap).

2. User painting/drawing tool for custom designs.
3. Option to export or share custom designs.

2.2.1.4 Try-On History Closet Module

1. Save the last five try-on sessions.
2. Display past tried shoes in a virtual “closet.”
3. Reuse cached try-ons without reloading.
4. Option to re-apply previously customized skins.

2.2.2 Additional Features

2.2.2.1 Multi-Shoe Comparison Module

1. Side-by-side comparison of two shoe models.
2. Left-vs-right foot shoe visualization.
3. Split-screen mode for multiple AR try-ons.

2.2.2.2 Save & Share Module

1. Capture screenshots of AR try-ons.
2. Share via social platforms and messaging apps.
3. Add watermarks for branding and authenticity.

2.2.2.3 Mix & Match Module

1. Combine different socks and shoe styles.
2. Preview outfit variations in AR.
3. Save favorite combinations.

2.2.2.4 Gesture Control Module

1. Using gestures to navigate the application interface.
2. Browsing the shoe catalog through gesture-based scrolling or swiping.
3. Selecting and initiating virtual try-on of shoes with predefined gesture commands.

To provide a clearer understanding of how these components interact, a feature diagram of the proposed system is presented in Figure 2.1. The diagram illustrates the relationship between the core modules and additional features, highlighting how they collectively

contribute to the overall functionality of the AR footwear try-on application. It visually summarizes the modular structure, showing how each element—such as AR try-on, outfit matching, shoe customization, and history management—integrates to create an interactive and user-friendly experience.

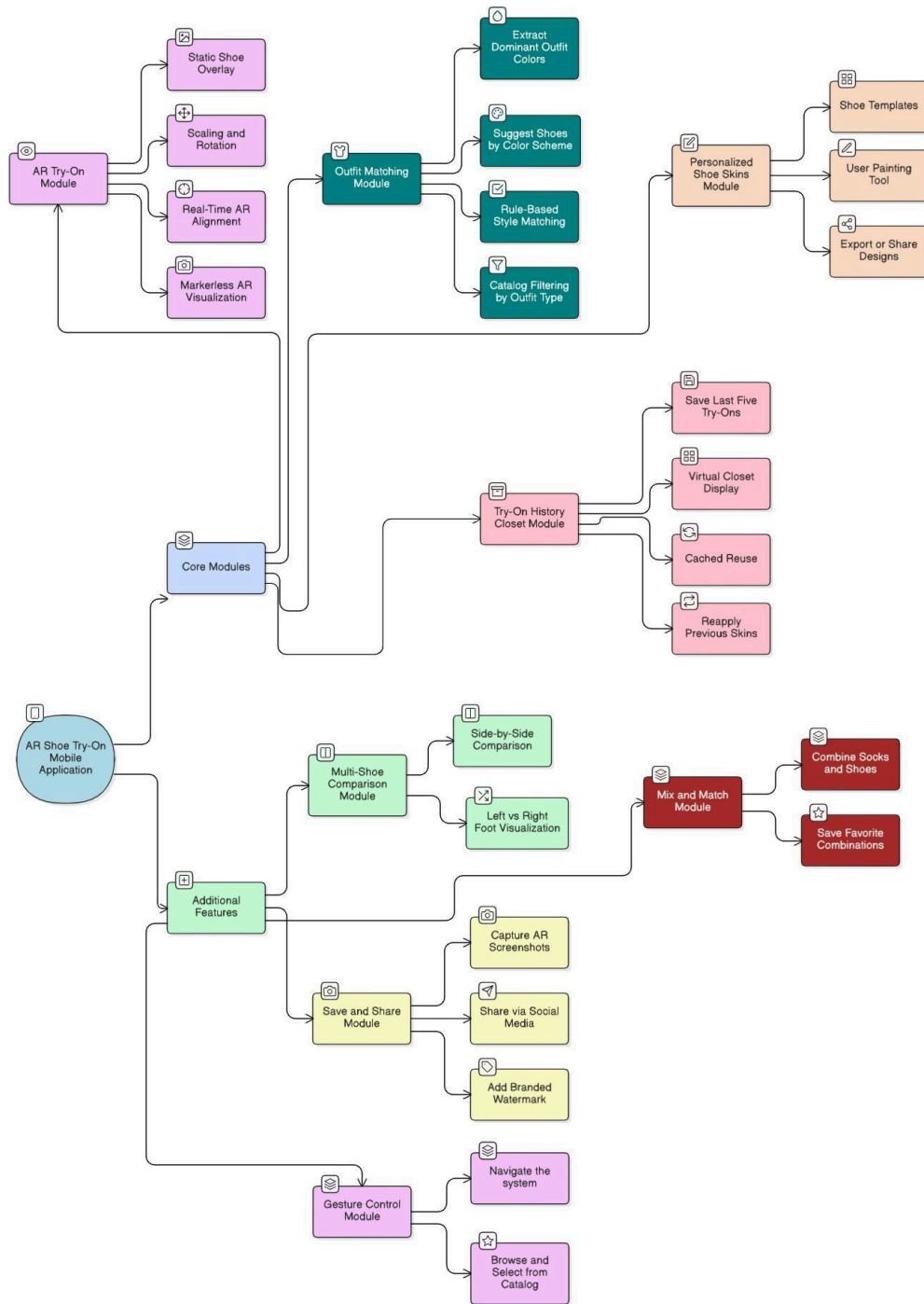


Figure 2.1: Feature diagram of the proposed AR footwear try-on system.

2.3 Tools and Technologies

The successful development of the proposed AR-based shoe try-on application relies on a diverse set of tools and technologies that span multiple domains, including augmented reality, computer vision, mobile development, 3D modeling, and data management. These technologies are carefully selected to ensure the system is both functional and scalable, while remaining practical within the constraints of a Final Year Project. The following categories outline the primary tools and technologies to be employed:

2.3.1 Augmented Reality Frameworks

The foundation of the application is built upon augmented reality capabilities. Google's **ARCore** is selected as the primary AR framework, providing markerless tracking, motion detection, and plane recognition optimized for Android devices. This ensures realistic alignment of virtual shoe models with users' feet. Additionally, **Unity with AR Foundation** may be considered as an optional framework to explore cross-platform scalability beyond Android.

2.3.2 Computer Vision and Image Processing

To enable advanced features such as foot detection, outfit color extraction, and real-time alignment, the system leverages established computer vision libraries. **OpenCV** will be used for low-level image preprocessing and feature extraction, while **MediaPipe** provides pre-trained models for hand and foot tracking, significantly reducing development time. For machine learning-driven tasks, such as outfit recommendation and personalization, **TensorFlow Lite** will be employed to deploy lightweight models directly on mobile devices.

2.3.3 Mobile Application Development

The application will be developed using **Android Studio**, with Java or Kotlin as the primary programming language, to ensure seamless integration with ARCore and Android's native APIs. Although frameworks such as **React Native** or **Flutter** offer cross-platform development, they are considered secondary options due to the complexity of integrating advanced AR features in hybrid environments. Native development ensures maximum performance and compatibility with ARCore.

2.3.4 3D Modeling and Assets

Realistic 3D shoe models form a core component of the user experience. The open-source tool **Blender** will be utilized to design, texture, and optimize 3D shoe models for real-time rendering. For rapid prototyping, external repositories such as **Sketchfab** and **TurboSquid** will be used to source high-quality pre-built assets. These resources reduce modeling overhead and allow the team to focus on AR integration and user interaction.

2.3.5 Database and Storage

The system will require both cloud and local storage solutions to manage try-on history, preferences, and product catalogs. **Firestore** will serve as the cloud database, enabling real-time synchronization and scalability. For offline functionality and caching, **SQLite** will be used to store user data locally on devices. This hybrid approach ensures reliability and accessibility regardless of network availability.

2.3.6 Backend and Integration (Optional)

While the primary focus of this project is the mobile AR application, optional backend integration will be considered for scalability. A lightweight backend built with **Node.js** and **Express** may be developed to provide APIs for catalog management, recommendation services, or future integration with e-commerce platforms. **REST APIs** will serve as the communication layer for interoperability with third-party systems.

In summary, this combination of tools and technologies ensures that the proposed solution is not only functional but also robust, modular, and scalable. The chosen stack balances academic exploration of computer vision and AR with industry-ready tools for mobile application development, creating a solution that is both research-driven and practically viable.

2.4 Work Division

The responsibilities of each group member are distributed across different modules and features of the project. Table 2.1 summarizes the allocation of tasks.

Table 2.1: Work Division of Project Team Members

Name	Registration	Responsibilities / Module / Feature
Ahmad Tashfeen	22i-2490	• Development of AR Try-On Module (base AR overlays, scaling, and alignment). • Implementation of Multi-Shoe Comparison and Walk Animation Preview. • Optimization of AR rendering performance on Android devices. • Collaboration on system integration and testing.
Ahmad Tariq	22i-1534	• Development of Outfit Matching Module (color extraction, rule-based matching). • Implementation of Save & Share and Environmental Try-On features. • Support in designing shoe catalog database and integration. • Contribution to UI design and prototyping in Figma.
Umer Qureshi	22i-2578	• Development of Personalized Shoe Skins Module (drawing tools, texture mapping). • Implementation of Try-On History Closet and Mix & Match Socks + Shoes. • Handling of local storage encryption and data security. • Documentation, testing, and quality assurance.

2.5 Timeline

The project will be executed in four iterations, each aligned with academic evaluation milestones. Table 2.2 outlines the planned tasks and deliverables for each iteration.

Table 2.2: Project Timeline and Iteration Plan

Iteration	Timeline & Evaluation	Planned Tasks & Deliverables
Iteration 1	August – October (Mid Evaluation)	• Design application skeleton (frontend navigation and UI prototype in Figma). • Implement basic shoe catalog and selection screen. • Prototype AR Try-On Module with static shoe overlay on foot (no animations). • Ensure markerless AR setup using AR-Core.
Iteration 2	November – December (FYP I Final Evaluation)	• Develop Core AR Try-On Module with shoe scaling, rotation, and improved alignment. • Implement Multi-Shoe Comparison feature (left vs right foot view). • Enable Save & Share Try-On functionality. • Develop basic Outfit Matching Module (color extraction and shoe recommendations). • Prototype Basic Gesture Controls. • Finalize frontend with working navigation and shoe catalog integration.
Iteration 3	January – February (Pre-Job Fair Evaluation)	• Develop Mix & Match Socks + Shoes functionality. • Fully Implement Gesture Controls. • Implement simplified Try-On History Closet (cache of last 5 try-ons). • Upgrade Outfit Matching Module with rule-based outfit-type recommendations. • Develop Personalized Shoe Skins Module (painting on templates mapped to 3D models).

Iteration	Timeline & Evaluation	Planned Tasks & Deliverables
Iteration 4	March – April (Final Evaluation & Post Job Fair)	• Conduct system-wide testing and bug fix- ing. • Optimize AR rendering for smoother per- formance on mid-range devices. • Secure local storage for personalized skins (encryption and version control). • Finalize documentation, polish UI/UX, and prepare the project for final submis- sion.

2.6 Project Software Architecture Deep Dive

2.6.1. Architecture Standard Model Selection

Excellent — since you’ve now finalized your **architecture, components, data flow, and storage model** in detail, here’s the **updated and consolidated Architecture Section** rewritten to fully align with the **final system you’ve designed** (SoleMate hybrid cloud-native architecture).

This version merges your latest changes, removes redundancies, and upgrades the tone to a polished, FYP-ready report style.

Everything reflects your final design (Node.js BFF, Firebase Auth/Firestore, MongoDB Atlas, Cloud Storage, Cloud Functions, Redis, and Android client).

Recommended Hybrid Architecture — Layered (N-Tier) Core + Cloud-Integrated Extensions

1. Overview

The final architecture of *SoleMate* follows a **Hybrid Layered (N-Tier) model**, enhanced with selective **serverless and cloud-integrated components**.

It combines the clarity of a layered monolith for on-device logic with the scalability of lightweight microservices for cloud-side computation and data management.

This approach ensures:

-  **Low latency** for AR-critical operations handled locally.
-  **Scalable and maintainable** backend for storage, recommendations, and analytics.
-  **Secure, modular** design leveraging Firebase authentication and cloud-hosted data.

2. Architecture Composition

Layer	Key Components	Responsibilities
Presentation Layer (Client)	Android App (Kotlin, MVVM), ARCore Interface, Gesture & Tap Input	User interaction, AR visualization, initiating business operations via ViewModels.
Application / Business Logic	AR Try-On Engine, Outfit Matching Engine, Skin Designer, Try-On History Closet, Multi-Shoe	Core domain logic (alignment, scaling, color matching, preview

Layer	Comparison	rendering, catalog querying).
Service / Integration Layer	Node.js Backend-for-Frontend (BFF) , Firebase Auth (on AWS EC2), Redis Cache, Cloud Functions (AI Processing)	API aggregation, authentication validation, caching, AI/ML inference, and communication with data stores.
Data Layer	MongoDB Atlas, Cloud Storage (AWS S3 / Firebase Storage), Firebase Firestore	Persistent structured data, large binary asset storage, and session/user metadata management.

3. Cloud & Data Interactions

All client-initiated network requests flow through the **Node.js BFF**, which serves as the single point of coordination between the mobile app and backend systems.

- The **BFF** validates Firebase tokens, aggregates data from MongoDB Atlas, and retrieves binary assets from Cloud Storage via presigned URLs.
 - **Firebase Auth / Firestore** manages user authentication and active session data.
 - **MongoDB Atlas** persists structured textual and numeric data (product metadata, user profiles, try-on records, custom skin metadata, and asset URLs).
 - **Cloud Storage (AWS S3 / Firebase Storage)** houses all heavy assets — 3D models, textures, media, and trained AI model files — referenced by URLs stored in MongoDB.
 - **Redis Cache** temporarily stores volatile session data and catalog snapshots for faster access but is never used for permanent storage.
 - **Cloud Functions** or serverless jobs handle computationally intensive tasks such as image color extraction, recommendation generation, and thumbnail creation, accessing AI models directly from Cloud Storage.
-

4. Data Flow Summary

1. **User Login** → Firebase Auth validates credentials; session metadata stored in Firestore.
2. **AR Try-On / Outfit Matching** → Android app processes AR data locally; metadata requests sent to BFF.
3. **Catalog Queries & Skin Uploads** → Routed through BFF → MongoDB (for metadata) and Cloud Storage (for media).

4. **Recommendation Requests** → BFF invokes Cloud Function / AI Model → returns curated results.
 5. **Caching** → Redis holds temporary catalog and session info for the active user.
 6. **Backups** → MongoDB Atlas backed up automatically to Cloud Storage for redundancy.
-

5. Key Architectural Constraints

- **Redis is volatile** — only used to cache data fetched from MongoDB/Firestore during active sessions.
 - **All user-uploaded content** (images, custom skins, AR snapshots) is written **directly to MongoDB + Cloud Storage** at upload time, never stored temporarily in Redis.
 - **AI Models** reside on Cloud Storage and are accessed by the Service Layer (not through MongoDB).
 - **Multi-Shoe Comparison** module receives direct non-gesture input (taps) from the UI to avoid control conflicts in dual-AR rendering mode.
-

6. Advantages of the Hybrid Model

- ✓ **Modularity & Testability** – Clear separation of UI, business logic, and backend services enables focused development and testing.
 - ✓ **Low Latency for AR** – Device-side processing minimizes round-trip delays for tracking and rendering.
 - ✓ **Scalability** – Serverless components and CDN-backed Cloud Storage scale automatically with demand.
 - ✓ **Security** – Firebase handles authentication and token-based authorization; presigned URLs secure asset downloads.
 - ✓ **Maintainability** – BFF simplifies client communication by aggregating APIs and shielding the mobile app from backend schema changes.
 - ✓ **Extensibility** – Individual services (e.g., recommendation or analytics) can evolve into standalone microservices when user scale increases.
-

7. Final Verdict

The finalized SoleMate system adopts a **Hybrid Layered Architecture** — combining a modular Android client with a **Node.js BFF**, **Firebase Authentication**, **MongoDB Atlas**, **Redis (volatile cache)**, and **serverless cloud functions** for AI workloads.

This design delivers **high performance, modularity, and scalability** without unnecessary DevOps overhead, ensuring a production-ready yet academically rigorous architecture suitable for AR-driven

mobile applications.

Architecture Diagram, Layout, and Iteration Plan

1) High-Level Hybrid Architecture

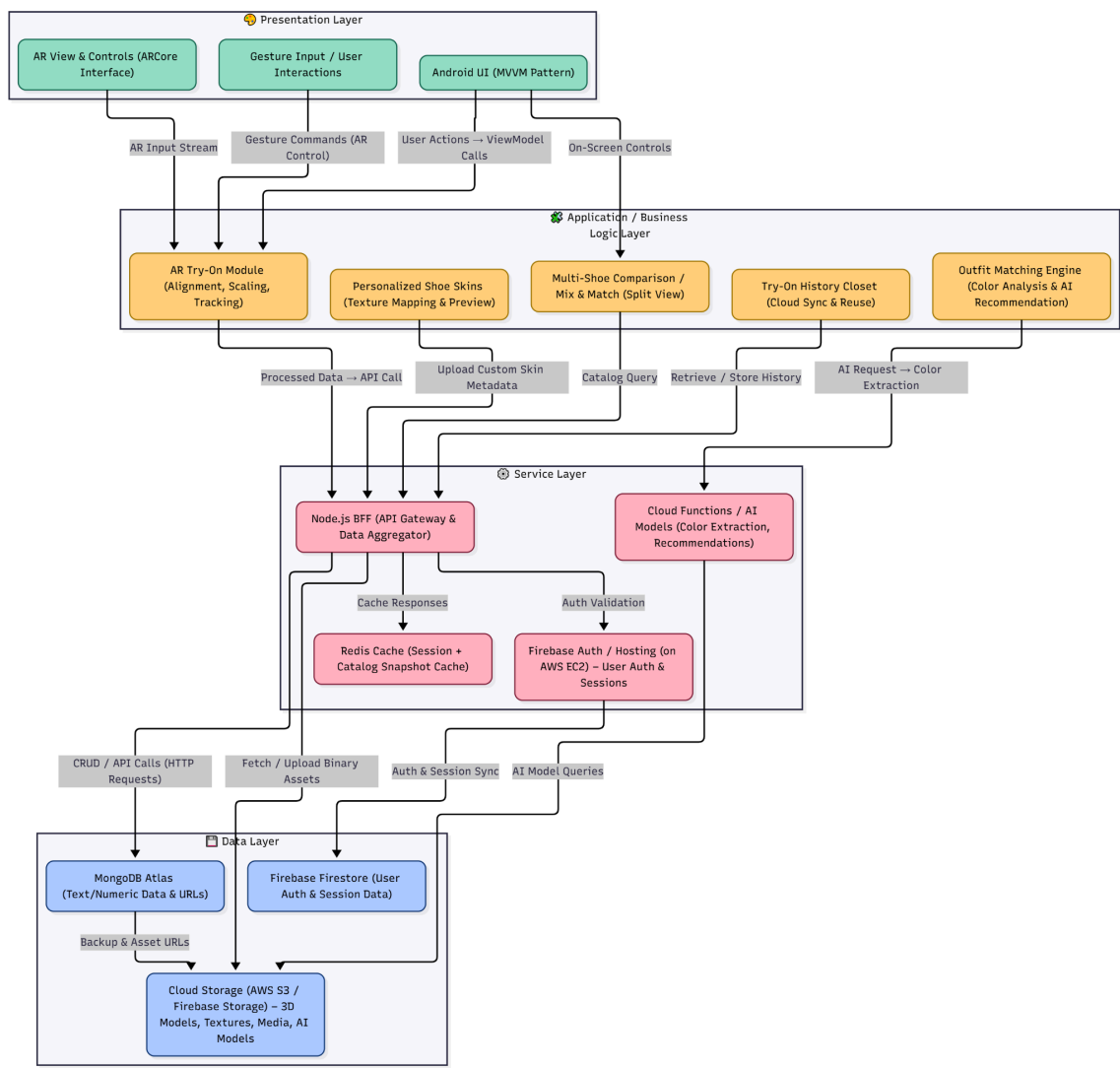


Figure 2. Hybrid Architecture Overview

Use the Mermaid code provided earlier or design an equivalent Visio / Draw.io diagram.

2) Gradle Module and Package Structure

```
RootProject/  
├─ app/
```

- └─ core/
- └─ feature-ar/
- └─ feature-catalog/
- └─ feature-outfit/
- └─ feature-skins/
- └─ feature-closet/
- └─ data-local/
- └─ data-remote/
- └─ test/

Module	Description
app	UI & navigation
core	Domain models & utils
feature-ar	ARCore integration
feature-catalog	Catalog management
feature-outfit	Outfit image processing
feature-skins	Skin customization
feature-closet	Try-on history
data-local	Room + DAOs
data-remote	Firebase/REST
test	Mocks & shared tests

3) Iteration 1 → 2 Implementation Checklist

Iteration	Goals	Tasks	Owner(s)	Acceptance Criteria
Iteration 1 (Aug–Oct)	App skeleton + Static AR prototype	Setup modules, UI skeleton, catalog, basic AR overlay, local cache	All (M1, M2, M3)	Project builds, static try-on works, catalog visible
		Implement DeviceProfile and cache assets	M3	Device info stored
		Unit tests + demo	All	CI build passes
Iteration 2 (Nov–Dec)	Dynamic AR Try-On, Multi-shoe comparison, Outfit matching	Improve AR tracking, LOD logic, caching	M1	Shoes stay aligned, ≥30 FPS

	Outfit color extraction (K-Means)	M2	3 dominant colors, ≤5s runtime
	Save & Share	M2, M3	Watermarked image saved & shared
	Remote catalog integration (BFF or Firebase)	M3	Metadata + presigned URLs fetched
	End-to-end performance tests	All	Stable FPS, accurate recommendations

Acceptance Examples

Feature	Test	Expected Result
AR Alignment	User moves device slowly	Shoe remains aligned (<5% drift)
Multi-Shoe Comparison	Load shoe A & B	Swap latency ≤ 2s
Outfit Matching	Upload image	3 recommendations shown
Save/Share	Capture AR view	Image saved, share intent opens

Developer Workflow

- Each feature module → has fake repository for local testing
 - GitHub Actions → builds, lints, runs tests on PR
 - Physical devices → mid-range + high-end for AR validation
-

Component Mapping (Summary)

Component	Responsibility
UI / ViewModel	Presentation and lifecycle
UseCases (Domain)	Core business logic
Repositories	Abstract storage + remote APIs
AR Module	Rendering, ARCore integration
CDN / Cloud Functions	Asset hosting and processing

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